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PERSONAL DATA

NATIONALITY AND BIRTH: Italy | 1996
PHD: 09/2024
CURRENTLY: Postdoctoral Researcher at Durham University (IPPP), UK
PREVIOUSLY: PhD Student at IFT UAM/CSIC Madrid (2021 - 2024), ES
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WEB OF SCIENCE: [JDW-8706-2023](#)

RESEARCH INTERESTS

Theoretical Particle Physics — BSM Phenomenology.

My research focuses on the phenomenology of physics beyond the Standard Model, with emphasis on extra-dimensional scenarios and their cosmological and astrophysical signatures. A central theme is the Kaluza-Klein (KK) gravitational sector: its spin-2, vector, and scalar towers, the sum rules and symmetry structures governing their interactions, and their interplay with dark matter, axions, and neutrinos across collider, astrophysical, and cosmological scales.

Keywords: *BSM, Extra Dimensions, Kaluza-Klein Phenomenology, Massive Spin-2, Dark Matter, Axions/ALPs, Neutrinos, Gravitational Waves.*

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EDUCATION & ACADEMIC STAYS

Education

Ph.D. in Physics , Instituto de Física Teórica (IFT UAM/CSIC), Madrid Final Grade: <i>Cum Laude</i> , International Mention; Advisor: Prof. Dr. Luca Merlo Thesis: <i>The Phenomenological Landscape of Leptophilic ALPs</i>	09/2021–09/2024
M.Sc. in Physics , LMU Munich Final Grade: 1.02/1.00; Advisors: Dr. G. Raffelt and S. Vogl Thesis: <i>Dark Matter Production in Warped Extra-Dimensions</i>	10/2018–12/2020
B.Sc. in Physics , University of Padova Final Grade: 109/110; Advisor: Prof. Dr. P. Marchetti Thesis: <i>Non-Abelian Anyons and Quantum Computation</i>	09/2015–07/2018

Academic Stays (2+ Months)

5. VISITOR VIA QUPIP GRANT at QUP-KEK , Tsukuba, Japan, in the group of <i>Dr. V. Takhistov</i>	04–06/2025
4. VISITOR at Ruprecht-Karls-Universität Heidelberg , in the group of <i>Prof. Dr. J. Jaeckel</i>	10–12/2023
3. VISITOR at Harvard University , Boston, in the group of <i>Prof. Dr. C. Argüelles-Delgado</i>	06–08/2023
2. RESEARCH INTERNSHIP at ETH , Zurich, in the group of <i>Prof. Dr. M.K. Marinkovic</i>	05–08/2021
1. STUDENT RESEARCH ASSISTANT at Max Planck Institut für Physik , Munich, in the group of <i>Dr. Habil. G. Raffelt</i>	01–11/2020

GRANTS, PRIZES & HONORS

Prizes and Awards

09/2025	2025 Prize in Theoretical Physics “Giuliano Preparata” Awarded by the Italian Physical Society (SIF). <i>“For significant contributions to theoretical and phenomenological aspects of physics beyond the Standard Model, in particular on new low-mass particles and extra dimensions.”</i>
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Grants

04/2025	QUP INTERNSHIP PROGRAM (QUPIP) International Center for Studies of the Universe and Particles (WPI-QUP), KEK, Tsukuba, Japan. See: QUPIP .
09/2021	DOCTORAL GRANT — European Union’s Horizon 2020, MARIE SKŁODOWSKA-CURIE grant agreement No 860881-HIDDEN, as Early Stage Researcher (ESR). See: hiddeneu.eu .

Minor Awards

- 1ST PRIZE in Poster Competition at INVISIBLE24 SCHOOL 07/2024
- 1ST PRIZE in Poster Competition at INVISIBLE23 SCHOOL 09/2023
- BEST POSTER PRESENTATION in Poster Competition at INVISIBLE22 SCHOOL 07/2022
- SCHOLARSHIP for high school students “PREMIO GELATI” (€1,000) 03/2016

COMMUNITY & LEADERSHIP

Collaborations

1. COST Action CA21106 COSMIC WISPERs:
 - Co-Leader Working Group 1 (Theory) 04/2025 – Today
 - Grants Evaluation Committee 09/2024 – Today
 - Young Researcher Council Representative 07/2024 – 05/2025
 - Monthly Colloquium Organiser 01/2023 – 07/2024

Editorial Experience

- Journal Referee for the *Journal of High Energy Physics (JHEP)* 06/2026 – Today
- Guest Editor for *Physics Letters B (PLB)* 03/2026 – Today
- Journal Referee for the *European Physical Journal C (EPJC)* 03/2025 – Today
- Editor for Proceedings of the 2nd Training School of COST Action COSMIC WISPerS (COSMICWISPerS2024) 06/2024

Organization of International Events

- International Organizing Committee, CA21106 4TH GENERAL MEETING 06/2026
CORFU SUMMER INSTITUTE, Corfu, Greece
- Junior Local Organizing Committee, EXTENDED WORKSHOP NUTs 2022 05/2022
INSTITUTE FOR THEORETICAL PHYSICS (IFT), Madrid, Spain

TALKS AT CONFERENCES & WORKSHOPS

Plenary Sessions

8. ALPs Across Scales, MITP - Mainz Institute for Theoretical Physics, Germany
Hunting Multiple Axions 06/2026
7. 5th Asian-European-Institutes Workshop for BSM, Durham University, UK.
New BSM transient signals from black hole formation 09/2025
6. Evolving Beyond the SM, CERN, Geneva, Switzerland.
New BSM transient signals from black hole formation 07/2025
5. QUP Workshop, KEK, Tsukuba, Japan.
Star shearing season: transient signals from black hole formation 04/2025
4. Invisibles24 Workshop, University of Bologna, Italy.
Extra-Dimensional Marion't 07/2024
3. Invisibles23 Workshop, University of Göttingen, Germany.
Down to the Seesaw Line via the JALZ ALP-HNL Portal 09/2023
2. 57th Rencontres de Moriond, La Thuille, Italy. Session: Electroweak Interactions & Unified Theories.

The Low Seesaw Scale Solution for M_W and $(g - 2)_\mu$ 03/2023

1. **Invisibles22 Workshop**, IJCLab, Orsay, France.
BSM flavoured correlations 06/2022

Parallel Sessions

5. **Pascos 2025**, Durham University, UK.
Star shearing season: transient signals from black hole formation 07/2025
4. **21st Lomonosov Conference**, Moscow State University, Russia.
The low-scale seesaw solution to the M_W and $(g - 2)_\mu$ anomalies 08/2023
3. **Planck 2023**, Warsaw, Poland.
BSM and 2HDM: update, prospects and a bridge to ALPs 05/2023
2. **DESY Theory Workshop**, DESY, Hamburg, Germany.
Spin-2 mediated Dark Matter in Warped Extra-Dimensions 09/2021
1. **2021 Phenomenology Symposium**, University of Pittsburgh, USA.
Spin-2 mediated Dark Matter in Warped Extra-Dimensions 05/2021

INVITED SEMINARS

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16. *The Hunt for Multiple Axions*. IFT, Madrid, July, 2026
 15. *The Hunt for Multiple Axions*. DESY, Germany, May, 2026
 14. *Hunting Multiple Axions*. Tsung-Dao Lee Institute, China, Mar., 2026
 13. *Hunting for Multiple Axions*. University of Basel, Switzerland, Feb., 2026
 12. *Transient Signals from Black Hole formation*. KEK Theory Center, Japan, June, 2025
 11. *Extra-dimensional axion patterns*. KEK Theory Center, Japan, Apr., 2025
 10. *The Extradimensional Portal to Dark Matter*. Institute of Cosmos Sciences of the University of Barcelona (ICCUB), May, 2024
 9. *The Minimal Massive Majoron*. Institute for Theoretical Physics, University Heidelberg, Dec., 2023
 8. *KK-Gravitons Sum Rules and Phenomenology*. University of Zurich, Nov., 2023
 7. *Flavour and Higgs physics in Z_2 -symmetric 2HD models near the decoupling limit*. LHC Higgs Working Group - WG2 WG3 joint meeting on CP violation in extended Higgs sector, Sept., 2023
 6. *Down to the Seesaw Line via the JALZ ALP-HNL Portal*. Harvard University, June, 2023
 5. *An old and a new opportunity for ALP physics*. Karlsruher Institut für Technologie, June, 2023
 4. *An Opportunity in the ALP-HNLs Sector*. COST Cosmic WISPer - WG1 Meeting, May, 2023
 3. *A Piece Beyond HNLs*. Albert-Ludwigs-Universität Freiburg, May, 2023
 2. *BSM Higgs Flavoured Correlation*. LHC Higgs Working Group - Common WG2 and WG3 – CP violation and Higgs Sector, June, 2022
 1. *Spin-2 mediated Dark Matter in Warped Extra-Dimensions*. Albert-Ludwigs-Universität Freiburg, July, 2021

TEACHING

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- **Tutor** at the 2026 STFC HEP Summer School 09/2026
(UK national PhD training school in HEP)

- **Tutor** at the 2025 STFC HEP Summer School (UK national PhD training school in HEP) 09/2025
- **Tutor** for Level 1 course, DURHAM UNIVERSITY FOUNDATIONS OF PHYSICS 1 (*40 hours*) 10/2025–04/2026
- **Tutor** for Level 1 course, DURHAM UNIVERSITY FOUNDATIONS OF PHYSICS 1 (*40 hours*) 10/2024–04/2025
- **Tutor** for Master course, LUDWIG-MAXIMILIANS-UNIVERSITÄT MÜNCHEN CLASSICAL AND QUANTUM SIMULATIONS OF PHYSICAL SYSTEMS, held by *Prof. Dr. M.K. Marinkovic* 10–12/2020
- **Tutor** for Master course, LUDWIG-MAXIMILIANS-UNIVERSITÄT MÜNCHEN INTRODUCTION TO LATTICE GAUGE THEORIES, held by *Prof. Dr. M.K. Marinkovic* 07–09/2020
- **Tutor** for Master course, LUDWIG-MAXIMILIANS-UNIVERSITÄT MÜNCHEN QUANTUM MECHANICS II, held by *Prof. Dr. V. Mukhanov* 10/2019–03/2020

PUBLICATIONS

I have authored 21 papers published (or in progress to be) in peer-reviewed journals, 4 community white papers, and 3 proceedings, with several further works in preparation. My research has been recognised by the 2025 SIF Prize in Theoretical Physics.

Publications

21. **A. de Giorgi**, J. Ingoldby, V. V. Khoze, and J. Turner, *Thermal Metastable Strings in One-Scale Models and Gravitational Waves*, [arXiv:2606.02689](#)
20. **A. de Giorgi**, D. Naredo-Tuero, and X. Ponce Díaz, *The Majoron Cosmological Window: Dark Matter and Thermal Leptogenesis*, [arXiv:2605.18944](#)
19. **A. de Giorgi**, Y. G. del Castillo, and J. Jaeckel, *Scalar Tsunamis from Black Hole Formation*, [arXiv:2602.06112](#)
18. **A. de Giorgi**, J. Jaeckel, S. Monath, and V. Takhistov, *Multiple Axions in Laboratory Experiments*, [arXiv:2512.16837](#)
17. **A. de Giorgi**, I. M. Soler, and S. Sevilano Muñoz, *Neutrino Constraints on Scalar-Tensor Gravity*, [arXiv:2512.13798](#)
16. **A. de Giorgi**, D. Pasari, and J. Turner, *Do neutrinos dream in 5D? Towards a comprehensive extra-dimensional neutrino phenomenology*, JHEP **05** (2026) 152, [[arXiv:2512.02101](#)]
15. **A. de Giorgi** and M. Ramos, *Extra-dimensional axion patterns*, Phys. Rev. D **111** (2025), no. 7 075006, [[arXiv:2412.00179](#)]
14. **A. de Giorgi** and J. Jaeckel, *Star shearing season: transient signals in wave-like dark matter experiments from black hole formation*, Eur. Phys. J. C **85** (2025), no. 2 155, [[arXiv:2408.10296](#)]
13. M. B. Marcos, **A. de Giorgi**, L. Merlo, and J.-L. Tastet, *ALPs and HNLs at LHC and muon colliders: Uncovering new couplings and signals*, SciPost Phys. **18** (2025) 084, [[arXiv:2407.14970](#)]
12. **A. de Giorgi**, M. F. Zamoro, and L. Merlo, *Visible GeV ALP from TeV Vector-Like Leptons*, Fortsch. Phys. **73** (2025), no. 3 2400165, [[arXiv:2402.14059](#)]
11. **A. de Giorgi** and S. Vogl, *Gravity-matter sum rules in models with a single extra-dimension*, JHEP **05** (2024) 315, [[arXiv:2311.01507](#)]

10. **A. de Giorgi**, L. Merlo, X. Ponce Díaz, and S. Rigolin, *The minimal massive Majoron Seesaw Model*, JHEP **03** (2024) 094, [[arXiv:2312.13417](#)]
9. **A. de Giorgi**, F. Koutroulis, L. Merlo, and S. Pokorski, *Flavour and Higgs physics in Z_2 -symmetric 2HD models near the decoupling limit*, Nucl. Phys. B **994** (2023) 116323, [[arXiv:2304.10560](#)]
8. **A. de Giorgi**, L. Merlo, and J.-L. Tastet, *Probing HNL-ALP couplings at colliders*, Fortsch. Phys. **71** (2023), no. 4-5 2300027, [[arXiv:2212.11290](#)]
7. **A. de Giorgi** and G. Piazza, *A lesson from $R_{\tau\tau}^{K^{(*)}}$ and $R_{\nu\nu}^{K^{(*)}}$ at Belle II*, Fortsch. Phys. (2024) [[arXiv:2211.05595](#)]
6. **A. de Giorgi**, L. Merlo, and S. Pokorski, *The Low-Scale Seesaw Solution to the M_W and $(g-2)_\mu$ Anomalies*, Fortsch. Phys. **71** (2023), no. 4-5 2300020, [[arXiv:2211.03797](#)]
5. J. Bonilla, **A. de Giorgi**, B. Gavela, L. Merlo, and M. Ramos, *The cost of an ALP solution to the neutral B-anomalies*, JHEP **02** (2023) 138, [[arXiv:2209.11247](#)]
4. **A. de Giorgi** and S. Vogl, *Warm dark matter from a gravitational freeze-in in extra dimensions*, JHEP **04** (2023) 032, [[arXiv:2208.03153](#)]
3. J. Alonso-Gonzalez, **A. de Giorgi**, L. Merlo, and S. Pokorski, *Searching for BSM physics in Yukawa couplings and flavour symmetries*, JHEP **05** (2022) 041, [[arXiv:2109.07490](#)]
2. **A. de Giorgi** and S. Vogl, *Dark matter interacting via a massive spin-2 mediator in warped extra-dimensions*, JHEP **11** (2021) 036, [[arXiv:2105.06794](#)]
1. **A. de Giorgi** and S. Vogl, *Unitarity in KK-graviton production: A case study in warped extra-dimensions*, JHEP **04** (2021) 143, [[arXiv:2012.09672](#)]

White Papers and Community Inputs

4. A. Arza *et. al.*, *The COSMIC WISPer White Paper: The physics case for Weakly Interacting Slim Particles*, [arXiv:2603.03433](#)
3. C. Albertus *et. al.*, *WISPeria – the WISPs Encyclopedia*, [arXiv:2602.09089](#)
2. D. Aybas *et. al.*, *Exploring the Dark Universe: A European Strategy for Axions and other WISPs Discovery*, [arXiv:2503.20432](#)
1. J.-H. Arling *et. al.*, *Early Career Researcher Input to the European Strategy for Particle Physics Update: White Paper*, [arXiv:2503.19862](#)

Proceedings

3. **A. de Giorgi**, M. Fuentes Zamoro, and L. Merlo, *GeV ALP from TeV Vector-like leptons*, in *Contribution to 2nd Training School and General Meeting of the COST Action COSMIC WISPer (CA21106) (COSMICWISPer2024) 10-14 June 2024 and 3-6 September 2024 Ljubljana (Slovenia) and Istanbul (Turkey)*, 11, 2024. [arXiv:2411.18246](#)
2. **A. de Giorgi**, L. Merlo, and S. Pokorski, *The Low-Scale Seesaw Solution to the M_W and $(g-2)_\mu$ Anomalies*, Moscow Univ. Phys. Bull. **79** (2024), no. Suppl 1 465–469
1. **A. de Giorgi**, L. Merlo, and S. Pokorski, *Low Seesaw Scale Solution for M_W and $(g-2)_\mu$* , in *57th Rencontres de Moriond on Electroweak Interactions and Unified Theories*, 4, 2023. [arXiv:2304.08438](#)

OUTREACH

6. **Podcast** on Particle Physics hosted by Dr. Shounak Bhattacharya 06/2025
Currently viewed 970+ TIMES. [Youtube Link](#)

5. **Public Lecture** “*Why neutrinos matter*” for the INTERNATIONAL MASTERCLASS IN PARTICLE PHYSICS 2025, Durham University, hosting over 110 STUDENTS from six local high schools. [Read more.](#) 03/2025
4. **Podcast** on Particle Physics hosted by Dr. Shounak Bhattacharya 01/2025
Currently viewed 520+ TIMES. [Youtube Link](#)
3. **Speaker**, JORNADAS DE LA FÍSICA, Parque de Atracciones de Madrid, ~1200 STUDENTS 03/2024
2. **Speaker**, JORNADAS DE LA FÍSICA, Parque de Atracciones de Madrid, ~1200 STUDENTS 03/2023
1. **A. de Giorgi**, *Physics, following the breadcrumbs of our intuition*, February, 2022

Non-Scientific Activities

May 2023 – Today	MEMBER, Rotaract Madrid Serrano. Community service organisation focused on international service projects.
May 2023 – Today	MENTOR, Lead the Future . Non-profit STEM mentorship organisation based on the spirit of ‘Give Back’.
Sep 2022 – May 2023	MEMBER, IFT Equity, Diversity and Inclusion Committee . EDI@IFT
2014 – 2021	FOUNDER AND CAO, Artupia (art-tech startup).
2012 – 2015	MEMBER AND EVENTS COORDINATOR, Europeanclubeuropeo . Club enhancing awareness of EU values of peace and cooperation among nations.

Languages and Computer Skills

Languages

Native: Italian | **Fluent:** English, Spanish | **Basic:** German

Computer Skills

Basic Knowledge: HTML, CSS
Intermediate Knowledge: C++, FeynCalc, FeynRules, xAct, FLAVIO
Advanced Knowledge: Mathematica, L^AT_EX, PYTHON

Arturo de Giorgi

4 June 2026